Data 227 - Autumn 2026 HW 4

Data Visualization and Communication - Trimble

Due: Thursday October 28, 2023 11:59pm

Misleading visualization design pair

1. For this assignment, you will create two static visualizations of a dataset of your choice: a visualization engineered to obscure some truth of the data; and a visualization with an "honest" presentation of the same data.

   All of the tools in our toolbox can be used well or poorly, and practicing using them to mislead will make us more savvy interpreters of data. Some datasets that may lend themselves to deception:

   (a) Centers for Medicare and Medicaid Open Payments database (pharma compensation to prescribers, dubbed by Pro publica "Dollars for Docs")

   (b) Guttmacher Institute Public-Use Datasets on Abortion and Fertility by Geography and Year

   (c) Indicators of Gender Equality from the World Bank 1960-2017

   (d) Customs and Border Protection Public Data Portal – travel stastics by time and place, law enforcement encouters by time, nationality, and legal authority

   Your obfuscation should be free of catastrophic flaws (like axes that have no meaningful labels) and should be technically accurate (don't fake the data) but should lead readers to a conclusion about the data that is false. (Make visualizations, find a fact in the data, make visualization to highlight the fact; negate the fact, make a new visualization, goose the visualization to encourage the incorrect reading.) Your visualizaitons should encode more than a handful of numbers, at least eight per graph.

   Your project submission should include:

   (a) A figure caption for each graphic, which needs to include provenance information sufficient to identify and locate the dataset.

   (b) A paragraph describing the techniques used to obfuscate the data.

   Submit the graphs and captions as a PDF, and the relevant code to produce the graph in a separate file.